

## QUESTION 25

### Multi-Document AI Assistant with Question Answering

**Documents:** ai.txt • ml.txt • nlp.txt

FAISS • all-MiniLM-L6-v2 • google/flan-t5-small

## 1. Aim

To build a document-based AI assistant capable of searching multiple uploaded documents, retrieving relevant document sections, and generating answers to user questions using a language model.

## 2. Problem Statement

A single-document assistant is limited when information is distributed across several files. A multi-document system combines several documents into one searchable vector space while preserving the source document associated with each retrieved chunk.

## 3. Objectives

- Load multiple external documents.
- Preserve source filenames for retrieved chunks.
- Generate semantic embeddings.
- Create a unified FAISS vector database.
- Retrieve the Top-5 relevant chunks.
- Display source documents and similarity scores.
- Generate answers from the retrieved multi-document context.

## 4. System Architecture

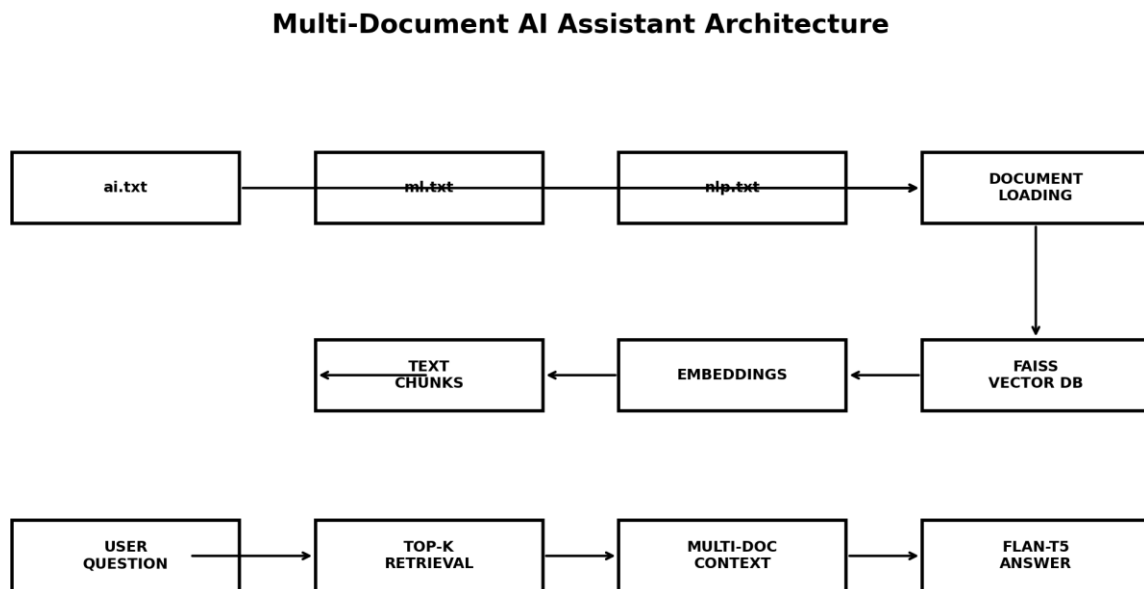


Figure 1: Multi-document AI assistant architecture.

## 5. Documents Used

| Document | Content Focus               |
|----------|-----------------------------|
| ai.txt   | Artificial Intelligence     |
| ml.txt   | Machine Learning            |
| nlp.txt  | Natural Language Processing |

## 6. Working Procedure

1. Load all TXT files from the documents folder.
2. Split each document into text chunks.
3. Store the source filename for every chunk.
4. Generate 384-dimensional embeddings using all-MiniLM-L6-v2.
5. Normalize embeddings and store them in FAISS IndexFlatIP.
6. Convert the user question into an embedding.
7. Retrieve the Top-5 most similar chunks across all documents.
8. Build a combined context from retrieved chunks.
9. Generate the final answer using google/flan-t5-small.

## 7. Technology Stack

| Component           | Actual Configuration |
|---------------------|----------------------|
| Documents Loaded    | 3                    |
| Embedding Model     | all-MiniLM-L6-v2     |
| Embedding Dimension | 384                  |
| Vector Database     | FAISS                |
| Index Type          | IndexFlatIP          |
| Top-K Retrieval     | 5                    |
| Generation Model    | google/flan-t5-small |

## 8. Actual Test Questions and Results

| Question                         | Top Source | Top Score | Generated Answer                                                                                                                  |
|----------------------------------|------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------|
| What is artificial intelligence? | ai.txt     | 0.8613    | a field of computer science that focuses on creating systems capable of performing tasks that normally require human intelligence |
| What is machine                  | ml.txt     | 0.8225    | a branch of artificial                                                                                                            |

|                                                               |         |        |                                                                                        |
|---------------------------------------------------------------|---------|--------|----------------------------------------------------------------------------------------|
| learning?                                                     |         |        | intelligence that enables computers to learn patterns from data                        |
| What is NLP?                                                  | nlp.txt | 0.7467 | Natural language processing                                                            |
| How are artificial intelligence and machine learning related? | ml.txt  | 0.7103 | a branch of artificial intelligence that enables computers to learn patterns from data |

## 9. Complete Retrieval Evidence

The implementation retrieved five sources for each query. The first query retrieved ai.txt first with 0.8613; the second retrieved ml.txt first with 0.8225; the third retrieved nlp.txt first with 0.7467. The cross-document relationship query retrieved ml.txt first with 0.7103 and ai.txt second with 0.6772.

## 10. Retrieval Score Analysis

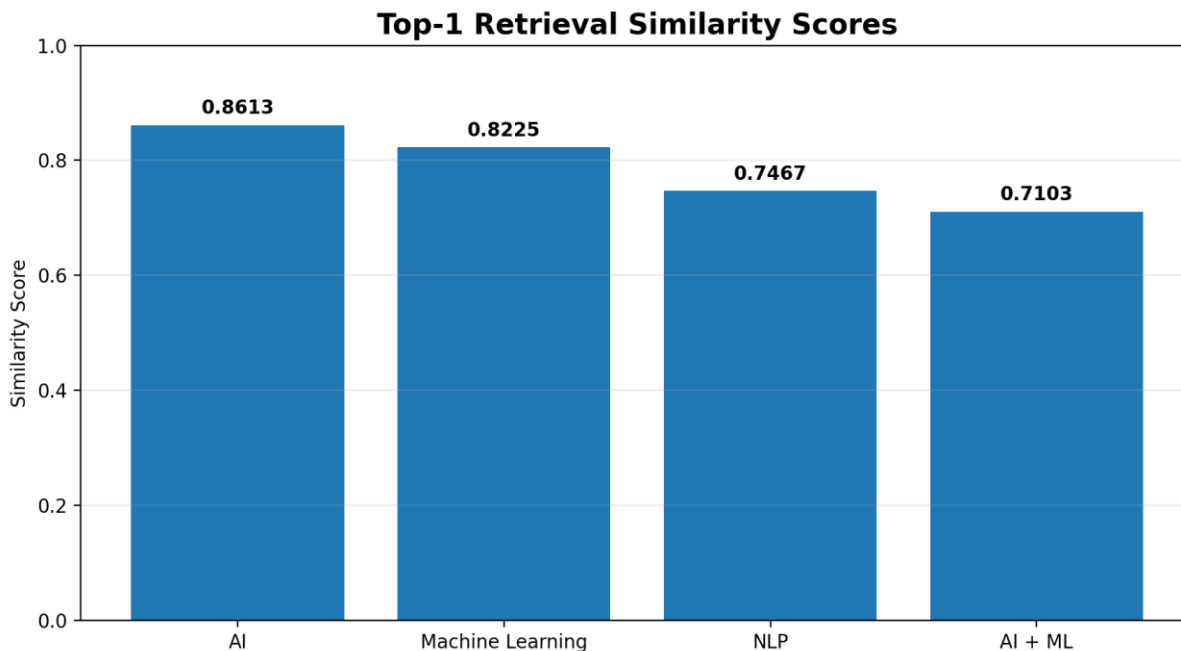


Figure 2: Top-1 similarity scores for the four test queries.

## 11. Cross-Document Retrieval Demonstration

For the question “How are artificial intelligence and machine learning related?”, the system retrieved ml.txt as the first source with a similarity score of 0.7103 and ai.txt as the second source with 0.6772. This is the key evidence that the assistant can search across multiple uploaded documents rather than being restricted to a single file.

## 12. Answer Analysis

The assistant produced relevant answers for the Artificial Intelligence, Machine Learning, and NLP questions. The relationship question returned the machine-learning definition, which is relevant but does not fully explain the relationship. This is a limitation of the small FLAN-T5 generation model rather than a failure of multi-document retrieval.

## 13. Result

The multi-document AI assistant successfully loaded three documents, created a unified vector database, retrieved Top-5 results, displayed source filenames and similarity scores, and generated answers. The highest observed Top-1 similarity score was 0.8613.

## 14. Advantages

- Supports multiple documents in one searchable knowledge base.
- Preserves source-document information.
- Can retrieve information from different files for one question.
- New documents can be added to the documents folder.
- Provides similarity scores for retrieval transparency.

## 15. Applications

- Multi-document question answering
- Research assistants
- Enterprise document assistants
- Educational knowledge systems
- Technical documentation search
- Organizational knowledge bases

## 16. Limitations

- Current implementation uses TXT files.
- The small FLAN-T5 model can generate incomplete relationship answers.
- Top-K retrieval may include less-relevant supporting chunks.

## 17. Conclusion

A multi-document AI assistant was successfully implemented using ai.txt, ml.txt, and nlp.txt. The system generated semantic embeddings, stored all document chunks in a unified FAISS vector database, retrieved the Top-5 relevant sections, identified their source documents, and generated answers using FLAN-T5. The cross-document query retrieved information from both ml.txt and ai.txt, confirming successful multi-document semantic retrieval.